

- 3 (a)** (thermal) energy/heat required to convert unit mass/1 kg of solid to liquid **M1**
 with no change in temperature/at melting point **A1 [2]**
- (b) (i)** energy required to warm ice = $24 \times 10^{-3} \times 2.1 \times 10^3 \times 15$ (= 756 J) **C1**
 energy required to melt ice at 0 °C = $24 \times 10^{-3} \times 330 \times 10^3$ (= 7920 J) **C1**
 total energy = 8700 J **A1 [3]**
- (ii)** energy lost by warm water = $200 \times 10^{-3} \times 4.2 \times 10^3 \times (28 - T)$ **C1**
 $200 \times 4.2 \times (28 - T) = 24 \times 4.2 \times T + 8676$ **C1**
 $T = 16 \text{ °C}$ **A1 [3]**
 [allow 2 marks if ΔT calculated]
 [allow 2 marks if $(24 \times 4.2 \times T)$ omitted]
 [allow 1 mark for $224 \times 4.2 \times (28 - T) = 8676$, $T = 19 \text{ °C}$]

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4. (a) acceleration proportional to displacement (from a fixed point)

M1